2.05 Trigonometry: Modelling in Radians

Worked Solutions 1–10

1 a)

$2\pi \div 1 = 2\pi$

1 b)

$2\pi \div 4\pi = \frac{1}{2}$

1 c)

$\pi \div \pi = 1$

1 d)

$2\pi \div \frac{\pi}{2} = 4$

1 e)

$2\pi \div \frac{\pi}{4} = 8$

1 f)

$\pi \div \frac{\pi}{2} = 2$

1 g)

$2\pi \div \frac{2\pi}{52} = 52$

1 h)

$2\pi \div \frac{2\pi}{7} = 7$

1 i)

$2\pi \div \frac{2\pi}{365} = 365$

1 j)

$2\pi \div \frac{\pi}{26} = 52$

2 a)

$y = \sin\left(\frac{2\pi}{7}x\right)$

2 b)

$y = \sin\left(\frac{2\pi}{1}x\right) = \sin(2\pi x)$

2 c)

$y = \sin\left(\frac{2\pi}{600}x\right) = \sin\left(\frac{\pi}{300}x\right)$

2 d)

$y = \sin\left(\frac{2\pi}{24}x\right) = \sin\left(\frac{\pi}{12}x\right)$

3 a)

$y = 3\sin\left(\frac{2\pi}{13}x\right)$

3 b)

$y = 2\sin\left(\frac{2\pi}{1}x\right) = 2\sin(2\pi x)$

3 c)

$y = 10\sin\left(\frac{2\pi}{20}x\right) = 10\sin\left(\frac{\pi}{10}x\right)$

3 d)

$y = -5\sin\left(\frac{2\pi}{30}x\right) = -5\sin\left(\frac{\pi}{15}x\right)$

4 a)

1

4 b)

1

4 c)

12

4 d)

2

5 a)

Max is 1, min is -1

5 b)

Max is $3 + 1 = 4$, min is $3 - 1 = 2$

5 c)

Max is $14 + 6 = 20$, min is $14 - 6 = 8$

5 d)

Max is $2 + 7 = 9$, min is $2 - 7 = -5$

6 a)

Period: 10

Amplitude: $\frac{5-1}{2} = 2$

Translate $\begin{pmatrix}0\\3\end{pmatrix}$

$$y = 3 + 2\sin\left(\frac{2\pi}{10}x\right) = 3 + 2\sin\left(\frac{\pi}{5}x\right)$$

6 b)

Period: 30

Amplitude: $\frac{2.5-1.5}{2} = \frac{1}{2}$

Translate $\begin{pmatrix}0\\2\end{pmatrix}$

$$y = 2 + \tfrac{1}{2}\sin\left(\frac{2\pi}{30}x\right) = 2 + \tfrac{1}{2}\sin\left(\frac{\pi}{15}x\right)$$

6 c)

Period: 20

Amplitude: $\frac{2-(-4)}{2} = 3$

Translate $\begin{pmatrix}0\\-1\end{pmatrix}$

$$y = -1 + 3\sin\left(\frac{2\pi}{20}x\right) = -1 + 3\sin\left(\frac{\pi}{10}x\right)$$

6 d)

Period: 7

Amplitude: $\frac{25-15}{2} = 5$

Translate $\begin{pmatrix}0\\20\end{pmatrix}$

$$y = 20 - 5\sin\left(\frac{2\pi}{7}x\right)$$

7 a)

Max – min $= 3.6 - 2 = 1.6$

$$\frac{1.6}{2} = 0.8$$

$$d = 2.8 + 0.8\cos\left(\frac{2}{5}\pi t\right)$$

Therefore $a = 2.8$, $b = 0.8$

7 b)

Period is $\frac{2\pi}{\frac{2}{5}\pi} = 5$

7 c)

$$2.8 + 0.8\cos\left(\frac{2}{5}\pi t\right) = 2.5$$

Find the first value:

$$\frac{2}{5}\pi t = \arccos(-0.375) = 1.95519310129\ldots$$

Find the second value:

$$2\pi - 1.95519310129\ldots = 4.32799220589\ldots$$
$$\frac{2}{5}\pi t = 4.32799220589\ldots$$

So $t = 1.556$ and $t = 3.444$

Add on the period: $1.556 + 5 = 6.556$

Time below 2.5 metres:

$$(3.444 - 1.556) + (7 - 6.556) = 2.332$$

2 hours 20 minutes

8 a)

$T = 36.8 + 0.64 = 37.44°\text{C}$

8 b)

First time $\sin\left(\frac{\pi}{12}(h+2)\right) = -1$ is:

$$\frac{\pi}{12}(h+2) = \frac{3\pi}{2}$$
$$h + 2 = 18 \Rightarrow h = 16 \text{ hours}$$

Therefore 16:00

8 c)

Max – min is: $39.2 - 38.3 = 0.9$

$$\frac{0.9}{2} = 0.45, \qquad 38.3 + 0.45 = 38.75$$
$$T = 38.75 - 0.45\sin\left(\frac{\pi}{12}(h+2)\right)$$

9 a)

Stretch parallel to t-axis, scale factor 2.4

Translate $\begin{pmatrix} 0 \\ 18.5 \end{pmatrix}$

9 b)

$x = 14$

$$t = 18.5 + 2.4\sin\left(\frac{2\pi}{52}(14+26)\right) = 16.1174987\ldots$$

Time of sunset is 16:07

10 a)

$$N = 140 + 120\sin\left(\frac{\pi}{6}(5-3)\right) = 243.923\ldots$$
$$N = 140 + 120\sin\left(\frac{\pi}{6}(9-3)\right) = 140$$

$243.923\ldots - 140 = 104$ passengers

10 b)

$$140 + 120 \sin\left(\frac{\pi}{6}(t-3)\right) = 200$$

Find the first value:

$$\frac{\pi}{6}(t-3) = \arcsin(0.5) = \frac{\pi}{6}$$

$t = 4$

Find the second value: $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$t = 8$

The period is: $2\pi \div \frac{\pi}{6} = 12$

Add on the period: $t = 4 + 12 = 16$, $t = 8 + 12 = 20$

Times at which 200 passengers: 04:00, 08:00, 16:00, 20:00

04:00 to 08:00, and 16:00 to 20:00

10 c)

The number of passengers is a discrete number which is then being represented by a continuous curve. It also assumes the same pattern outside of the data collection times, so for example, the model says the platform is never empty and that there are symmetries around the times when there are a maximum number of passengers, so it predicts the same number of passengers at both 2am and 10am which seems unrealistic!